



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

● Experiment No.: 4

1. **Title:** Use function types, scope, and closures; apply try-catch; build a palindrome checker.

2. **Software/Tools Required:** VS Code, JavaScript.

3. Theory:

Anonymous Function: A function without a name, assigned to a variable and used to display the welcome message.

Global Scope: Variables declared outside functions (e.g., language) are accessible throughout the program.

Local Scope: Variables declared inside a function (e.g., topic) are accessible only within that function.

Closure: A function that retains access to its outer function's variables even after the outer function has finished execution. The counter() function demonstrates this by preserving the count variable.

Arrow Function: A concise syntax for writing functions, used here to display output messages.

Try-Catch: Used for exception handling to catch and handle invalid input without stopping program execution.

Palindrome Checker: Compares the original string with its reversed version. If both are equal, the string is a palindrome; otherwise, it is not.

4. Output :

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
pratham@Prathams-MacBook-Air-2 jsexp4 % node "/Users/shreyamaske/jsexp4/shreya1196prac4.js"
=====
Shreya's 1196 Palindrome Checker Program
=====
Language: JavaScript
Topic: Functions
Count: 1
1221 is a palindrome.
```



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```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● pratham@Prathams-MacBook-Air-2 jsexp4 % node "/Users/shreyamaske/jsexp4/shreya1196prac4.js"
=====
Shreya's 1196 Palindrome Checker Program
=====
Language: JavaScript
Topic: Functions
Count: 1
Count: 2
abc1221cba is a palindrome.
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● pratham@Prathams-MacBook-Air-2 jsexp4 % node "/Users/shreyamaske/jsexp4/shreya1196prac4.js"
=====
Shreya's 1196 Palindrome Checker Program
=====
Language: JavaScript
Topic: Functions
Count: 1
daman is not a palindrome.
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● pratham@Prathams-MacBook-Air-2 jsexp4 % node "/Users/shreyamaske/jsexp4/shreya1196prac4.js"
=====
Shreya's 1196 Palindrome Checker Program
=====
Language: JavaScript
Topic: Functions
Count: 1
Count: 2
Error: Input must be a string.
```

5. Result/Conclusion:The program successfully demonstrates the use of JavaScript function types, variable scope, closures, arrow functions, and exception handling using try-catch. It correctly checks whether a given string is a palindrome while illustrating important JavaScript programming concepts in a simple and effective manner.



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Case Study No.: 4

- Case Study Title:** Real time case study ATM card pin verification using JavaScript functions case study an ATM system verifies whether a customers pin is a palindrome. If the pin reads the same forward and backward the Bank displays a special security message. To implement this, create a function to reverse the entered pin number and another function to check whether it is a palindrome. use function declaration, function expression, Arrow function and demonstrate scope and closure concepts.

Case Study Program Code:

```
<!DOCTYPE html>

<html>

<head>

    <title>ATM PIN Verification</title>

</head>

<body style="background-color: rgb(233, 166, 58);">

<h2 >Shreya's (1196) ATM PIN Verification System</h2>

Enter ATM PIN:

<input type="text" id="pin">

<button onclick="verifyPIN()">Verify PIN</button>

<hr>

<div id="output"></div>

<script src="shreya1196case4.js"></script>

</body>

</html>
```



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```
// Anonymous Function (Function Expression)
let welcome = function () {
    return "<h3>ATM PIN Verification System</h3>";
};

// Global Variable
let bankName = "Shreya's (1196) Bank";

// Function Declaration - Scope
function scopeDemo() {
    let branch = "Nagpur"; // Local Variable
    return "<b>Bank:</b> " + bankName +
        "<br><b>Branch:</b> " + branch + "<br>";
}

// Closure
function counter() {
    let count = 0;
    return function () {
        count++;
        return count;
    };
}

let c = counter();

// Function Declaration - Reverse PIN
function reversePIN(pin) {
    return pin.split("").reverse().join("");
}

// Function Declaration - Check Palindrome
function palindrome(pin) {
    let reverse = reversePIN(pin);
    return pin === reverse;
}
```



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```
// Arrow Function
const display = (message) => {
    document.getElementById("output").innerHTML +=
message + "<br>";
};

// Main Function
function verifyPIN() {
    document.getElementById("output").innerHTML =
"";
    display(welcome());
    display(scopeDemo());
    display("<b>Verification Attempt:</b> " + c());
    try {
        let pin =
document.getElementById("pin").value;
        if (pin === "") {
            throw "Please enter ATM PIN. ";
        }
        if (palindrome(pin)) {
            display("<h3 style='color:red'>Weak
Security!</h3>");
            display("PIN is a palindrome.");
            display("Choose a stronger PIN.");
        }
        else {
            display("<h3 style='color:green'>Strong
Security!</h3>");
            display("PIN is not a palindrome.");
        }
    }
    catch(error) {
        display("<b>Error:</b> " + error);
    }
}
```



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}

3. Output:

24070521196_Shreya Maske_ x

ATM PIN Verification x +

< > ↺

File /Users/shreyamaske/jsexp4/shreya1196case4.html

⌵

Shreya's (1196) ATM PIN Verification System

Enter ATM PIN:

ATM PIN Verification System

Bank: Shreya's (1196) Bank
Branch: Nagpur

Verification Attempt: 1

Strong Security!

PIN is not a palindrome.



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24070521196_Shreya Maske_ x

ATM PIN Verification x +

< > ↺

File /Users/shreyamaske/jsexp4/shreya1196case4.html

Shreya's (1196) ATM PIN Verification System

Enter ATM PIN:

ATM PIN Verification System

Bank: Shreya's (1196) Bank
Branch: Nagpur

Verification Attempt: 2

Weak Security!

PIN is a palindrome.
Choose a stronger PIN.

4. Result/Conclusion:

The ATM PIN verification system was successfully implemented using JavaScript. The program correctly identified whether the entered PIN was a palindrome, displayed "**Weak Security**" for palindrome PINs and "**Strong Security**" for non-palindrome PINs, while demonstrating function declaration, function expression, arrow function, scope, closures, and exception handling using try-catch.